

Ken (Xiaokang) Liu

Software Development Engineer

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Summary

Software Development Engineer with 4+ years of experience at AWS S3 building scalable distributed systems and data infrastructure. Proven ability to design high-performance services, optimize large-scale systems, and deliver significant cost savings (>\$49M). Strong background in API design, data processing, storage systems, and system reliability at scale.

Work Experience

Software Development Engineer

Seattle, WA

Amazon Web Services

Feb 2022 - Present

- Designed and built a **distributed data lifecycle management service** to clean up expired data copies used in checksum validation, improving storage efficiency and protecting upstream service profit margins
- Developed **scalable high-throughput backend APIs** for archival data deletion, eliminating orphaned data creation
- Built an automated orphaned data detection and deletion system, resulting in over **\$49 million** in CapEx savings
- Re-architected the service request message processing component with integrated cloud database solutions, reducing host CPU usage and message processing time by **90% (from seconds to milliseconds)**
- Improved system performance and memory efficiency, reducing **30%** heap memory usage and saving **\$760K** annually
- Upgraded the **Apache Spark** framework across EMR clusters, reducing runtime by **30%** and saving over **\$130K** annually
- Established a cross-team rapid archival data restoration service, reducing **Glacier Flexible Retrieval** restore times by up to **85% (from hours to minutes)** without additional infrastructure cost to customers
- Designed end-to-end system architectures for scalable data processing services, ensuring high availability, fault tolerance, and efficient resource utilization

Research Assistant

Newark, DE

University of Delaware

Feb 2019 - Feb 2022

- Developed algorithms for image reconstruction and solving inverse problems using MATLAB and Python
- Applied machine learning techniques (SVM, perceptron) on experimental data to classify image patterns
- Analyzed reconstruction quality and optimized the design of the Coded Aperture patterns with the algorithm developed

Skills

Languages	Java, Kotlin, Python, C/C++, Scala, TypeScript, Ruby
Backend & Systems	Distributed Systems, REST APIs, Microservices, CI/CD, Asynchronous Processing
Data & Infrastructure	Data Storage Systems, Apache Spark, Large-scale Data Processing
Cloud Platform	AWS (CloudFormation, DynamoDB, EBS, EC2, ECS, EMR, Lambda, S3, SQS)
Specialties	System Design, Performance Optimization, Scalability, Reliability Engineering

Publication

- Edgar Salazar, **Xiaokang Liu**, and Gonzalo Arce, "X-ray Compton backscattering imaging via structured light," Opt. Express 30, 15211-15226 (2022)
- Angela P. Cuadros, **Xiaokang Liu**, Paul E. Parsons, Xu Ma, and Gonzalo R. Arce, "Experimental demonstration and optimization of X-ray StaticCodeCT," Appl. Opt. 60, 9543-9552 (2021)

Education

Master of Science, Electrical & Computer Engineering

Newark, DE

University of Delaware

Feb 2019 - Feb 2022

Bachelor of Science, Electrical Engineering

Newark, DE

University of Delaware

Aug 2015 - Dec 2018